

Lucas Root

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Computer Science student (Cybersecurity & AI threads) at the Georgia Institute of Technology, seeking a software engineering internship.

EDUCATION

Georgia Institute of Technology <i>B.S. in Computer Science: Cybersecurity & AI</i>	Atlanta, GA Aug. 2026 - Expected May 2028
University of Kansas <i>B.S. in Computer Science & Cybersecurity Engineering, Honors College</i> — 3.9/4.0 GPA — Coursework: Programming 2, Software Engineering, Discrete Structures, Embedded Systems	Lawrence, KS Aug. 2024 - May 2026
Menlo-Atherton High School <i>High School Diploma, 4.08 GPA</i>	Menlo Park, CA May 2024

EXPERIENCE

AI/ML Research Intern <i>Air Force Research Laboratory (AFRL/RI) - Griffiss Institute - VICEROY MAVEN</i>	Jun. 2026 - Aug. 2026 Rome, NY
- Contributed to 5 privacy and AI research initiatives over an 8-week internship within the ZEIOS Vanguard group. SWEEP & SHADOW: Developed privacy-preserving retrieval mechanisms using Python, OHTTP/ODoH protocols, and mitmproxy to obfuscate client metadata during data queries. WHISPER: Benchmarked LLM-driven Private Information Retrieval (PIR) systems integrated with local LLMs to enable privacy-preserving queries against weather APIs. SCOPE: Performed video annotation and inter-annotator agreement analysis using Krippendorff's Alpha to ensure dataset quality for downstream vision tasks. RANGER: Evaluated real-time streaming video-language models (VLMs) under latency and continuous input constraints.	
Founder & Lead Developer, Joumé <i>Clothing brand - design, dev, sourcing, finances</i>	Dec. 2024 - Present Menlo Park, CA
Founded and run a clothing brand end-to-end: storefront (Rust, SQLite, Stripe), design, sourcing, and finances.	
Navy Midshipman <i>US Navy ROTC, University of Kansas</i>	Aug. 2024 - Dec. 2024 Lawrence, KS
Completed New Student Indoctrination and Orientation; recognized for leadership potential.	

PROJECTS

TURT: Custom Unix Shell  <i>C, Linux System Calls, Makefile</i>	Oct. 2025 - Present
- Engineered a POSIX-compliant shell in C, managing process execution via <code>fork</code>, <code>execvp</code>, and <code>wait</code>. - Built a custom tokenizer and signal handling for built-in commands, replicating core Bash functionality.	
Praxis: Student Social Network  <i>Rust, Next.js, Postgres</i>	Jan. 2026 - Present
- Architecting a full-stack platform connecting students for project collaboration, with a Rust backend for type safety and low-latency queries. - Designing a relational database schema in PostgreSQL to manage user relationships and project feeds.	
Syllabus2Cal  <i>TypeScript, Next.js, OpenAI API</i>	Jan. 2025 - Present
- Built a tool converting unstructured syllabus text into structured .ics data using LLM function calling. - Optimized API usage costs and response times through effective prompt engineering and caching.	

LEADERSHIP & AWARDS

President, Kansas Crew <i>University of Kansas</i>	Aug. 2024 - Present Lawrence, KS
Lead logistics, recruitment, and community partnerships for the official university rowing club.	
Varsity Rower, NorCal Crew <i>Redwood City, CA</i>	Aug. 2019 - May 2024
Competed nationally (13th at 2023 Head of the Charles); coached novice rowers ages 11-15.	
KU Excellence Scholarship <i>University of Kansas</i>	2024

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, Java, C#, JavaScript/TypeScript, SQL
Developer Tools: Git/GitHub, Docker, Linux/Unix, GDB, Make, VSCode
Concepts: Memory Management, OS Architecture, Embedded Systems, Data Structures